

Controllers Synthesis for Max-Plus Linear Systems under State Constraints

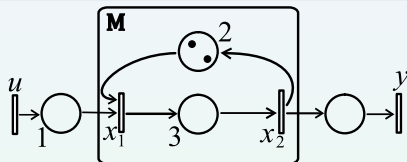
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Problem Statement



Preliminary

The max-plus algebraic framework is a powerful tool for modeling and analyzing discrete event systems such as Timed Event Graphs (TEGs).

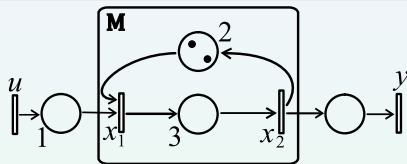
Questions ?

- Is it possible to control the system in order to ensure some state specifications as:
 - The delay between the firing of two transitions must not be greater than a given value,
 - The number of tokens in a place must not be smaller than a given value.

Outline

- 1 TEG Model in (max,plus) algebra
- 2 (max,plus) algebra in few words
- 3 Control Problem
- 4 Example

TEG Model in (max,plus) Algebra



Firing Date [Cohen et al., 85]

$x_i(k)$: date of the firing numbered k for the transition labelled x_i .

In conventional algebra

$$x_1(k) = \max(1 + u(k), 2 + x_2(k - 2))$$

$$x_2(k) = 3 + x_1(k)$$

$$y(k) = x_2(k)$$

In (max-plus) algebra

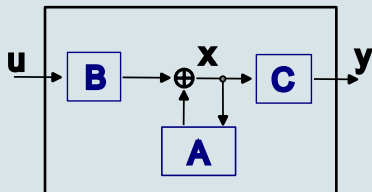
$$x_1(k) = 1 \otimes u(k) \oplus 2 \otimes x_2(k - 2)$$

$$x_2(k) = 3 \otimes x_1(k)$$

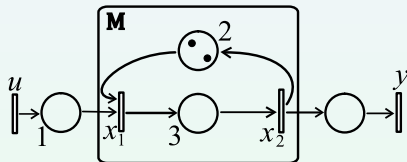
$$y(k) = x_2(k)$$

TEG Model in (max,plus) Algebra

Dynamic Model



$$\begin{aligned} x(k) &= Ax(k-1) \oplus Bu(k) \\ y(k) &= Cx(k) \end{aligned}$$



In the example above:

$$\begin{aligned} A_0 &= \begin{pmatrix} \epsilon & \epsilon \\ 3 & \epsilon \end{pmatrix}, A_2 = \begin{pmatrix} \epsilon & 2 \\ \epsilon & \epsilon \end{pmatrix} \\ B &= \begin{pmatrix} 1 \\ \epsilon \end{pmatrix}, C = (\epsilon \quad e) \end{aligned}$$

Idempotent Semiring

Definition [Baccelli et al., 1992], [Heidergott et al., 2006]

An Idempotent Semiring $(\mathcal{D}, \oplus, \otimes)$ is a set $\mathcal{D}$, equipped with two operations, where $\oplus$ is associative, commutative and admits ε as a neutral element, $\otimes$ is associative and admits e as a neutral element, $\otimes$ distributes over the $\oplus$, and ε is absorbing ($a \otimes \varepsilon = \varepsilon$).

Furthermore, the $\oplus$ is idempotent, i.e., $\forall a \in \mathcal{D}, a = a \oplus a$.

Set $\mathcal{D}$ is a lattice with ε as the bottom and $\top$ as the top element.

$$a \succeq b \Leftrightarrow a = a \oplus b \Leftrightarrow b = a \wedge b$$

Example (max,plus) Algebra

$\overline{\mathbb{Z}}_{\max} = (\{\mathbb{Z} \cup -\infty \cup +\infty\}, \max, +)$ is an idempotent semiring

- $\oplus = \max, \otimes = +, \varepsilon = -\infty, e = 0, \top = +\infty$
- $1 \otimes 1 = 2, 5 \oplus 6 = 6 \Leftrightarrow 5 \preceq 6 \Leftrightarrow 5 \wedge 6 = 5$ (classical order)

Residuation of mappings

Residuation Theory (Croisot 56, Blyth 05)

If $f(x) \preceq y$ admits a greatest solution, then f is residuated and $f^\#$ is the unique mapping s.t.: $f \circ f^\# \preceq Id$ and $f^\# \circ f \succeq Id$.

Inequality $a \otimes x \preceq b$ [Baccelli et al. 92]

Mapping $L_a : x \mapsto a \otimes x$ is residuated, and $L_a^\# : y \mapsto a \oslash y$, then inequality $a \otimes x \preceq b$ admits a greatest solution, denoted $x = a \oslash b$.

Example: $\overline{\mathbb{Z}}_{\max}$

Inequality $3 \otimes x \preceq 5$ admits a greatest solution $x = 3 \oslash 5 = 5 - 3 = 2$. It achieves equality in the scalar case.

Matrix

Sum of matrices

$$(A \oplus B)_{ij} = A_{ij} \oplus B_{ij}$$

$$\begin{pmatrix} 2 & 5 \\ 3 & 4 \end{pmatrix} \oplus \begin{pmatrix} e & 8 \\ 1 & 7 \end{pmatrix} = \begin{pmatrix} 2 & 8 \\ 3 & 7 \end{pmatrix}$$

Product of matrices

$$(A \otimes B)_{ik} = \bigoplus_{j=1 \dots n} (A_{ij} \otimes B_{jk})$$

$$\begin{pmatrix} 2 & 5 \\ \varepsilon & 3 \\ 1 & 8 \end{pmatrix} \otimes \begin{pmatrix} e \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \otimes e \oplus 5 \otimes 1 \\ \varepsilon \otimes e \oplus 3 \otimes 1 \\ 1 \otimes e \oplus 8 \otimes 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 4 \\ 9 \end{pmatrix}$$

Matrix

Residuation of matrices

$A \oslash B$ is the greatest solution of $A \otimes X \preceq B$

$$(A \oslash B)_{ik} = \bigwedge_{j=1 \dots n} (A_{ji} \oslash B_{jk})$$

Example

$$A \otimes X = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \otimes \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \preceq \begin{pmatrix} 8 \\ 9 \\ 10 \end{pmatrix} = B$$

$$A \oslash B = \begin{pmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{pmatrix} \oslash \begin{pmatrix} 8 \\ 9 \\ 10 \end{pmatrix} = \begin{pmatrix} (1 \oslash 8) \wedge (3 \oslash 9) \wedge (5 \oslash 10) \\ (2 \oslash 8) \wedge (4 \oslash 9) \wedge (6 \oslash 10) \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \end{pmatrix}$$

Dealing with equations

Fixed point equations

For order preserving mapping $f \preceq Id$, it is possible to compute fixed point $f(x) = x$ by considering an iterative algorithm.

Iterative Algorithm

Input x_0 , output $x = f(x) \preceq x_0$.

$x = x_0$

while $(x \neq f(x))$

$x = f(x)$

end

Application

By starting from an x_0 , a solution to $Dx = Ex$ can be found by considering mapping $f(x) = D \backslash (Ex) \wedge E \backslash (Dx) \wedge x$. This mapping is isotone and s.t. $f \preceq Id$.

Dealing with equations

Kleene Star operator *

$$a^* = \bigoplus_{i \in \mathbb{N}_0} a^i = e \oplus a \oplus a^2 \oplus \dots$$

Application

Over a complete idempotent semiring $\mathcal{D}$, the least solution to an implicit equation:

$$x = ax \oplus b \Leftrightarrow x = a^*b$$

In particular, the following equivalence holds:

$$x = ax \Leftrightarrow x = ax \oplus x \Leftrightarrow x = a^*x$$

Dealing with equations

γ transform [Cohen, Quadrat et al. IEEE TAC 89]

γ transform of $s(k)$ is a formal series $s(\gamma) = \bigoplus_{k \in \mathbb{N}} \gamma^k s(k)$.

γ -operator is like a backward shift operator in the event domain,
 $y(\gamma) = \gamma x(\gamma) \Leftrightarrow y(k) = x(k-1) \forall k$. The set of series is a semiring denoted $\mathbb{Z}_{\max}[[\gamma]]$.

Dynamic Model

$$x(k) = Ax(k-1) \oplus Bu(k)$$

$$y(k) = Cx(k)$$

$$\Leftrightarrow$$

$$x(\gamma) = A\gamma x(\gamma) \oplus Bu(\gamma)$$

$$y(\gamma) = Cx(\gamma)$$

Dealing with equations

Solving the implicit equation

$$x(k) = Ax(k-1) \oplus Bu(k) \Leftrightarrow x(k) = A^* Bu(k)$$

$$y(k) = Cx(k) \Leftrightarrow y(k) = CA^* Bu(k)$$

$$x(\gamma) = A\gamma x(\gamma) \oplus Bu(\gamma) \Leftrightarrow (A\gamma)^* Bu(\gamma)$$

$$y(\gamma) = Cx(\gamma) \Leftrightarrow C(A\gamma)^* Bu(\gamma)$$

Transfer matrix H

$$H = C(\hat{A}\gamma)^* B$$

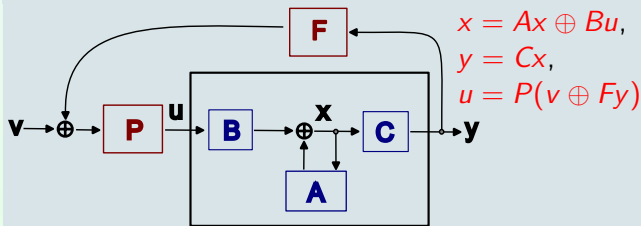
$$A = \hat{A}\gamma, \text{ so } H = CA^* B$$

Control problem

Problem definition

To find a neutral control law, $u = P(v \oplus Fy)$, for the system, to ensure that the state trajectories satisfy a two-sided equation $Nx = Mx$, where matrices $N, M \in (\mathbb{Z}_{\max}[\gamma])^{n \times n}$ characterize constraints between internal transitions of the TEG.

Dynamic Model with Controller



Control problem

Algebraic Development

In order to have a neutral controller, the new transfer matrix H for the controlled system must preserve the best system's input/output performance.

$$x = Ax \oplus BP(v \oplus Fy) \Leftrightarrow x = A^*BP(v \oplus Fy)$$

$$y = Cx = CA^*BPv \oplus CA^*BPFy \Leftrightarrow y = (CA^*BPF)^*CA^*BPv$$

$$H = (CA^*BPF)^*CA^*BP$$

First constraint relation:

$$(CA^*BPF)^*CA^*BP \preceq CA^*B$$

Control problem

Algebraic Development

The second constraint relation is the two-sided equation that characterizes constraints between internal transitions of the TEG.

$$x = A^*BP(v \oplus Fy), Nx = Mx$$

$$N(A^*BP(v \oplus Fy)) = M(A^*BP(v \oplus Fy))$$

Causality

To ensure the controller is realizable, it must be causal, that is:

$$Pr_+(F) = F$$

$$Pr_+(P) = P$$

Control problem - Methodology

Proposition: The greatest P

Let's first consider the open-loop problem ($F = \epsilon$), that is the computation of matrix P capable to solve the following equations:

$$(CA^*BPF)^*CA^*BP \preceq CA^*B \Leftrightarrow CA^*BP \preceq CA^*B$$

$$N(A^*BP(v \oplus Fy)) = M(A^*BP(v \oplus Fy)) \Leftrightarrow N(A^*BP) = M(A^*BP)$$

The first equation

► How?

The upper bound for the controller P can be found:

$$CA^*BP_{opt} = CA^*B \Leftrightarrow P_{opt} = (CA^*B) \oslash (CA^*B)$$

The second equation

► How?

$$f(P) = Pr_+(P) \wedge (NA^*B) \oslash (MA^*BP) \wedge (MA^*B) \oslash (NA^*BP)$$

Methodology

Consider mapping

and the iterative algorithm with $P_0 = P_{opt}$ in order to find the greatest $P \preceq P_0$ yielding a control that respects the constraints. The optimal solution P_{opt}^+ is the greatest such that:

$$P_{opt}^+ = Pr(P_{opt}^+)$$

Control problem

Proposition: The greatest F

Consider the closed-loop problem, with control law and state expression given as follows:

$$u = P_{opt}^+(v \oplus Fy)$$

$$x = A^* B P_{opt}^+ (F C A^* B P_{opt}^+)^* v$$

The objective is to find the optimal feedback F_{opt}^+ such that:

$$N A^B P_{opt}^+ (F_{opt}^+ C A^* B P_{opt}^+) = M A^B P_{opt}^+ (F_{opt}^+ C A^* B P_{opt}^+)$$

Control problem - Methodology

Methodology

► How?

As shown before, the controller has to be neutral:

$$CA^B P_{opt} (FCA^* B P_{opt}) \preceq CA^* B$$

$$P_{opt} (FCA^B P_{opt}) \preceq (CA^* B) \oslash (CA^* B) = P_{opt}$$

$$(FCA^B P_{opt}) \preceq P_{opt} \oslash P_{opt}$$

If $(FCA^B P_{opt}) \preceq P_{opt} \oslash P_{opt}$, also $(FCA^* B P_{opt})$ is.

$$FCA^* B P_{opt} \preceq P_{opt} \oslash P_{opt}$$

$$F \preceq P_{opt} \oslash P_{opt} \oslash (CA^* B P_{opt}) = F_{opt}$$

Control problem - Methodology

Methodology

Let us note that $\forall F$:

$$CA^B P_{opt}^+ (FCA^* BP_{opt}^+) \preceq CA^* BP_{opt}^+$$

$$CA^B \succeq CA^* BP_{opt}^+ (FCA^* BP_{opt}^+) \preceq CA^* BP_{opt}^+ \preceq CA^* B$$

$$CA^B P_{opt}^+ (FCA^* BP_{opt}^+) = CA^* BP_{opt}^+$$

So $F_{opt}^+ = Pr_+(P_{opt} \setminus P_{opt} \setminus (CA^* BP_{opt}))$ is the greatest solution.

Control problem

Main Result

Consider a system $x = Ax \oplus Bu$, $y = Cx$.

A neutral controller $u = P(v \oplus Fy)$ can be designed to satisfy the constraint $Nx = Mx$.

- 1 First, compute the initial parameter:

$$P_{opt} = (CA^*B) \oslash (CA^*B)$$

- 2 Apply an iterative algorithm with the mapping:

$$f(P) = Pr_+(P) \wedge (NA^*B) \oslash (MA^*BP) \wedge (MA^*B) \oslash (NA^*BP)$$

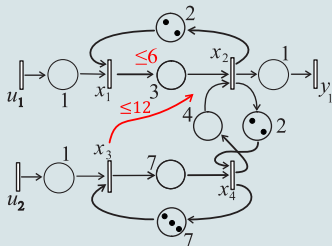
starting from $P_0 = P_{opt}$, to find the greatest P_{opt}^+ .

- 3 Then, compute the greatest:

$$F_{opt}^+ = Pr_+(P_{opt} \oslash P_{opt} \oslash (CA^*BP_{opt}))$$

Model and constraints

TEG model



Model of the system and set of constraints

- The model of this system is given by

$$x = \begin{pmatrix} \varepsilon & 2\gamma^2 & \varepsilon & \varepsilon \\ 3 & \varepsilon & \varepsilon & 4 \\ \varepsilon & \varepsilon & \varepsilon & 7\gamma^3 \\ \varepsilon & 2\gamma^2 & 7 & \varepsilon \end{pmatrix} x \oplus \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & \varepsilon \\ \varepsilon & 1 \\ \varepsilon & \varepsilon \end{pmatrix} u$$

$$y = (\varepsilon \quad 1 \quad \varepsilon \quad \varepsilon) x$$

- Maximal time between x_3 and x_2 is fixed to 12, and between x_1 and x_2 is fixed to 6:

$$N = \begin{pmatrix} e & \varepsilon & \varepsilon & \varepsilon \\ \varepsilon & e & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & e & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon & e \end{pmatrix} M = \begin{pmatrix} \varepsilon & -6 & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon & \varepsilon \\ \varepsilon & -12 & \varepsilon & \varepsilon \\ \varepsilon & \varepsilon & \varepsilon & \varepsilon \end{pmatrix}$$

After Computation

Numerical results

By applying the iterative algorithm, starting with $P_0 = P_{opt}$, in order to obtain P_{opt}^+ :

$$\begin{aligned} P_0 &= (CA^*B) \oslash (CA^*B) \\ &= \begin{pmatrix} 0\gamma^0 \oplus (6\gamma^2 \oplus 12\gamma^4)r^* & (8\gamma^0 \oplus 14\gamma^2)r^* \\ -16\gamma^0 \oplus (-10\gamma^1 \oplus -2\gamma^2)r^* & (0\gamma^0 \oplus 6\gamma^2)r^* \end{pmatrix} \end{aligned}$$

and

$$P_{opt}^+ = \begin{pmatrix} (12\gamma^4 \oplus 20\gamma^5)r^* & (8\gamma^0 \oplus 14\gamma^2)r^* \\ (4\gamma^4 \oplus 12\gamma^5)r^* & (0\gamma^0 \oplus 6\gamma^2)r^* \end{pmatrix}$$

where r is $(14\gamma^3)$.

After Computation

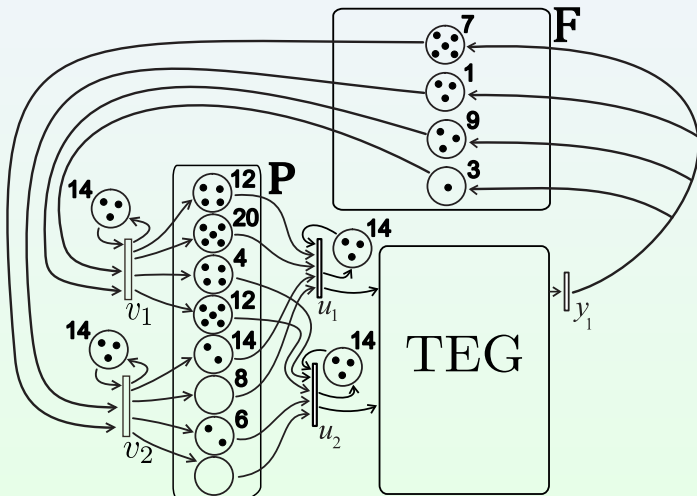
Numerical results

So we calculate $F_{opt}^+ = Pr_+(P_{opt} \setminus P_{opt} \setminus (CA^*BP_{opt}))$:

$$F_{opt}^+ = \begin{pmatrix} (3\gamma^1 \oplus 9\gamma^3)r^* \\ (1\gamma^3 \oplus 7\gamma^5)r^* \end{pmatrix}$$

where r is $(14\gamma^3)$.

Controlled TEG



Further Works

Further works

- When initial state is out of the semimodule of constraints, after some iterations the state vector will reach the semimodule, but we would like to give an upper bound for the number of iterations to reach this semimodule.

Software tools available for free

<https://github.com/santastico/CDC2025---Simulation>

Thanks for your attention !

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